



INDIGENE

STUDY CONSOLE

ANA 203

BASIC HISTOLOGY & CYTOLOGY

Complete Study Compendium

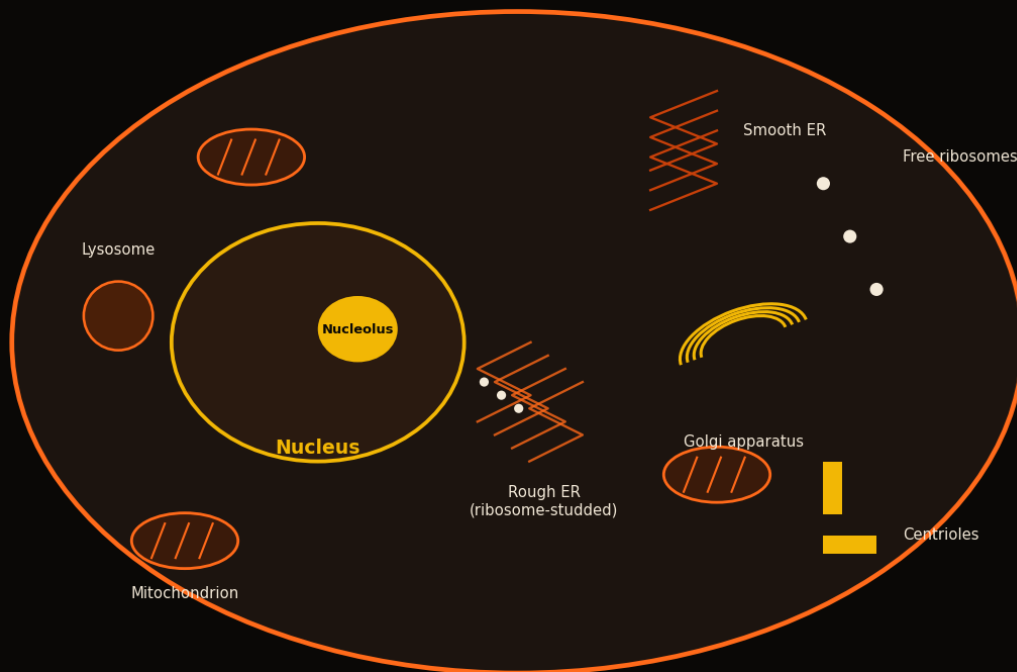
Federal University Oye-Ekiti | Nursing Science | 200 Level
The Cell • Membrane Transport • Cell Division • Epithelial Tissue
Connective Tissue • Muscle Tissue • Nerve Tissue • Blood • Cartilage • Bone

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1. The Cell

THE CELL — Generalized Structure



Overview

- The cell is the smallest structural and functional unit of life — every tissue in the body is built from cells and their products.
- Two broad regions: **Plasma membrane** (the boundary/gatekeeper) and the **cytoplasm** (organelles + cytosol) surrounding a **nucleus**.

Plasma (Cell) Membrane

- Described by the **Fluid Mosaic Model**: a phospholipid bilayer with proteins floating/embedded in it like tiles in a mosaic.
- Phospholipids are amphipathic — hydrophilic (water-loving) heads face outward/inward toward water; hydrophobic (water-fearing) tails face each other in the middle.
- Integral proteins span the membrane (e.g. channels, carriers); peripheral proteins sit on the surface (e.g. enzymes, receptors).
- Cholesterol is wedged between phospholipids — stabilizes fluidity, prevents the membrane from becoming too rigid (cold) or too fluid (heat).
- Glycoproteins and glycolipids on the outer surface form the **glycocalyx** — the cell's 'ID card' for recognition (blood typing, immune recognition).

■ ALZHEIMER'S MODE

Picture the membrane as a lively dance floor (fluid) covered with mosaic tiles (proteins) — heads of the dancers point out toward the crowd (water), tails huddle together avoiding the crowd (hydrophobic core). Cholesterol is the bouncer keeping the dance floor from getting too wild or too stiff.

Organelles

Organelle	Structure	Function
Nucleus	Double membrane (envelope) with pores; contains nucleolus + chromatin	Control centre — stores DNA, directs protein synthesis
Rough ER	ER studded with ribosomes	Synthesizes and folds proteins for export/membranes
Smooth ER	ER without ribosomes	Lipid synthesis, detox, calcium storage (muscle: sarcoplasmic reticulum)
Golgi apparatus	Stack of flattened membranous sacs (cisternae)	Modifies, packages, and ships proteins/lipids in vesicles
Mitochondria	Double membrane; inner folded into cristae; own DNA	Powerhouse — ATP production via aerobic respiration
Lysosomes	Membrane-bound sacs of hydrolytic enzymes	Intracellular digestion; 'suicide bags' in cell death
Peroxisomes	Small membrane-bound sacs	Break down fatty acids, detoxify hydrogen peroxide
Ribosomes	Free or ER-bound; rRNA + protein	Site of protein synthesis (translation)
Cytoskeleton	Microfilaments, microtubules, intermediate filaments	Shape, support, intracellular transport, movement
Centrioles	Paired cylindrical structures (9+0 triplets)	Organize the mitotic spindle during cell division

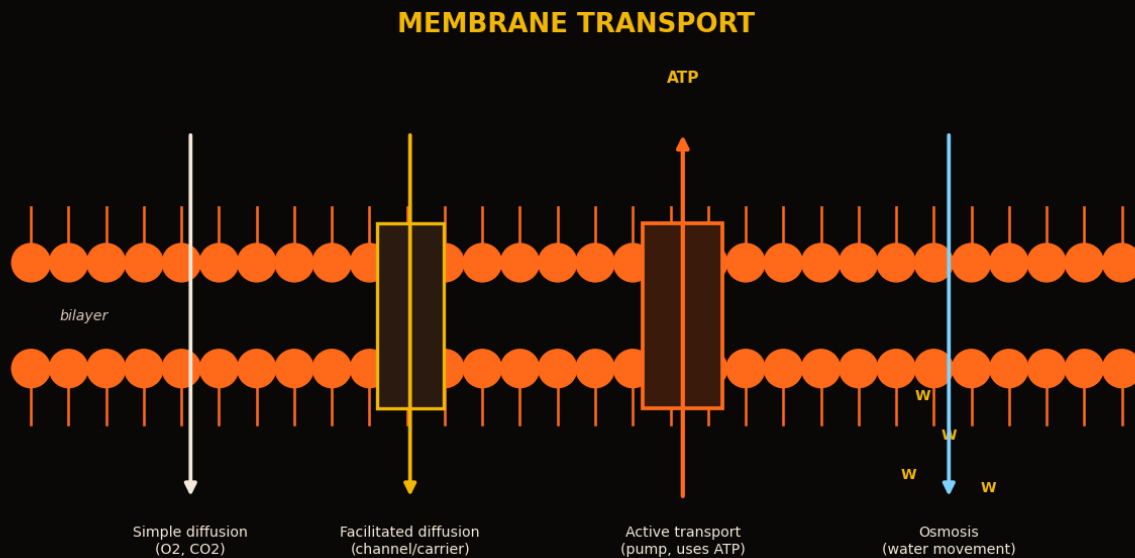
Nucleus (detail)

- **Nuclear envelope:** double membrane continuous with rough ER, perforated by nuclear pores controlling molecule traffic in/out.
- **Nucleolus:** dense region inside the nucleus where ribosomal RNA is made and ribosome subunits assembled.
- **Chromatin:** DNA + histone proteins; condenses into visible chromosomes only during cell division.

■ EXAM HACK — Organelle recall

'Rough' ER = Ribosomes = protein factory. 'Smooth' ER = no ribosomes = lipids/detox. Mitochondria has its OWN DNA — a classic trick-question trap (endosymbiotic theory). Lysosome = 'Lysis' = breakdown/digestion; remember as the cell's stomach/demolition crew.

2. Membrane Transport



Passive Transport (no ATP required — moves WITH the concentration gradient)

Type	Description	Example
Simple diffusion	Small/non-polar molecules move directly across the lipid bilayer, high → low concentration	O ₂ , CO ₂ , lipid-soluble hormones
Facilitated diffusion	Molecules move through a channel or carrier protein, high → low, still no ATP	Glucose via GLUT transporters, ions via channels
Osmosis	Diffusion of WATER across a selectively permeable membrane toward higher solute concentration	Water movement in/out of RBCs
Filtration	Movement driven by hydrostatic (pressure) gradient, not concentration	Fluid filtration at the glomerulus/capillaries

Active Transport (REQUIRES ATP — moves AGAINST the gradient)

- **Primary active transport:** ATP is used directly by a pump protein, e.g. the Na⁺/K⁺-ATPase pump (3 Na⁺ out, 2 K⁺ in per ATP).
- **Secondary active transport:** uses the energy stored in an ion gradient (created by a primary pump) to move another substance — *co-transport* (same direction, symport) or *counter-transport* (opposite direction, antiport), e.g. Na⁺/glucose symporter in the gut.
- **Bulk transport** (vesicular, also ATP-dependent): **Endocytosis** — cell engulfs material (phagocytosis = 'cell eating' solids; pinocytosis = 'cell drinking' fluids; receptor-mediated endocytosis = specific ligand-receptor uptake). **Exocytosis** — vesicles fuse with the membrane to release contents outside (e.g. hormone/neurotransmitter secretion).

■ ALZHEIMER'S MODE

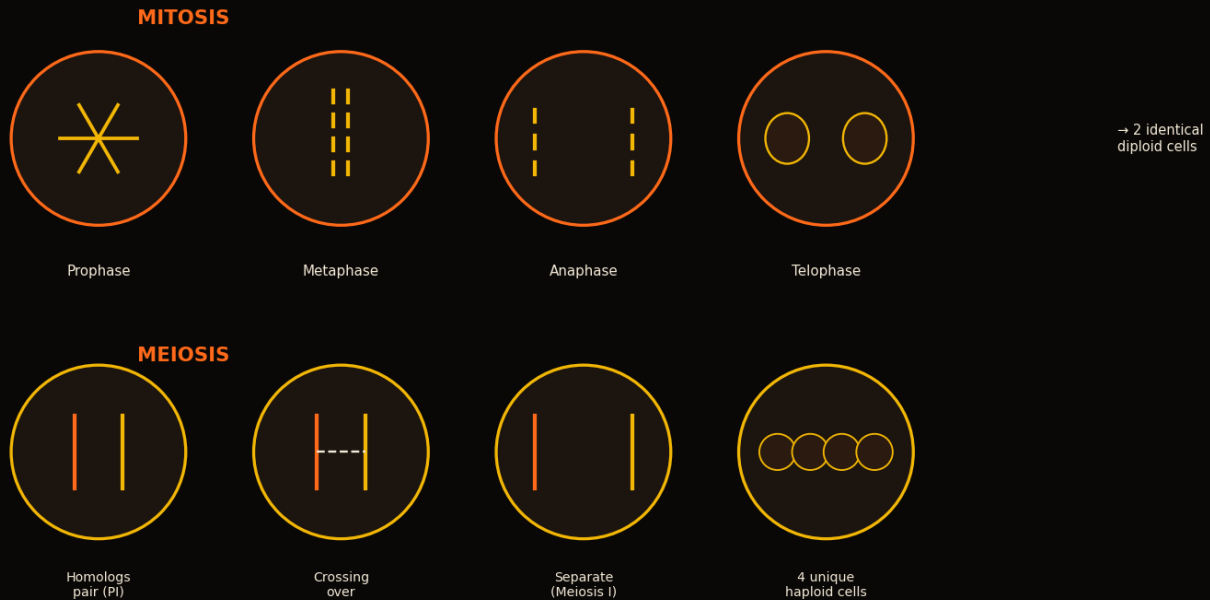
Passive transport is like water rolling downhill — no engine needed, gradient does the work. Active transport is like pushing a boulder uphill — you need fuel (ATP) because you're going against the natural flow.

■ EXAM HACK — Osmosis vs tonicity

Hypotonic solution → water rushes INTO the cell → cell swells/bursts (haemolysis in RBCs). Hypertonic solution → water rushes OUT → cell shrinks (crenation). Isotonic → no net water movement. Trap: 'facilitated diffusion needs ATP' is FALSE — it's still passive, just needs a protein escort, not fuel.

3. Cell Division — Mitosis & Meiosis

CELL DIVISION OVERVIEW



Mitosis (somatic cell division — growth & repair)

Phase	Key Event
Interphase (G1, S, G2)	Cell grows, DNA is replicated (S phase) — NOT technically part of mitosis proper
Prophase	Chromatin condenses into visible chromosomes (sister chromatids); spindle fibres begin forming; nuclear envelope breaks down
Metaphase	Chromosomes line up single-file at the metaphase plate (equator)
Anaphase	Sister chromatids are pulled apart to opposite poles by spindle fibres
Telophase	Chromosomes decondense; two new nuclear envelopes form
Cytokinesis	Cytoplasm divides — two genetically identical diploid (2n) daughter cells result

Meiosis (gamete formation — reduces chromosome number by half)

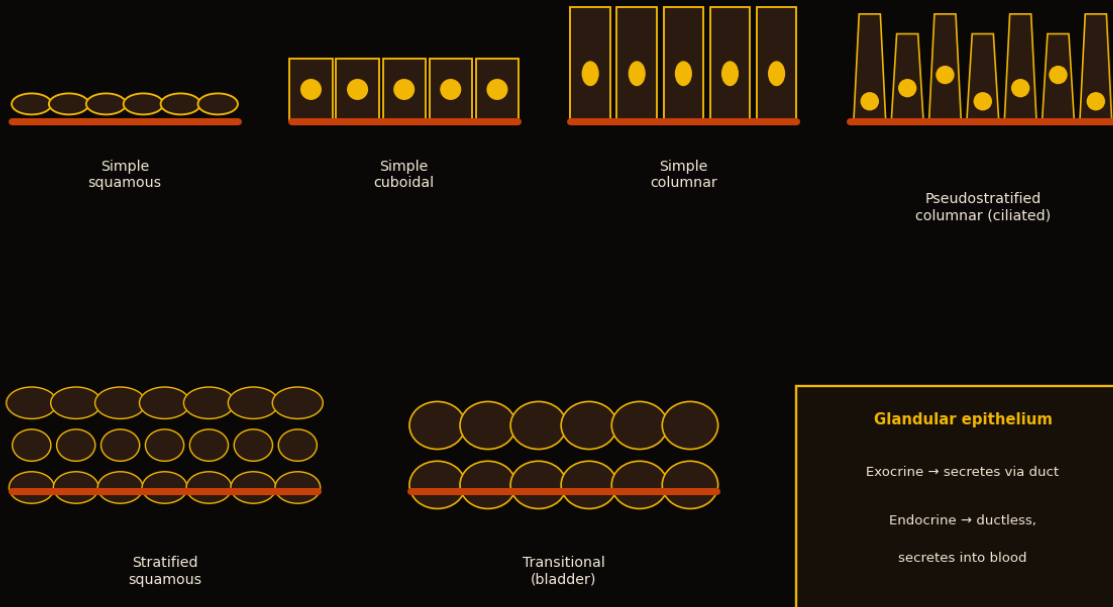
- Two successive divisions (Meiosis I and Meiosis II) from one diploid cell, producing four genetically unique haploid (n) daughter cells.
- **Meiosis I** (reductional division): homologous chromosomes pair up (synapsis) and exchange segments — **crossing over** — in Prophase I, then homologous pairs separate in Anaphase I.
- **Meiosis II** (equational division): mechanically like mitosis — sister chromatids finally separate.
- Purpose: genetic variation (via crossing over + independent assortment) and maintaining the correct chromosome number across generations.

Feature	Mitosis	Meiosis
Divisions	One	Two (I and II)

Feature	Mitosis	Meiosis
Daughter cells	2, diploid (2n), identical	4, haploid (n), genetically unique
Purpose	Growth, repair, asexual reproduction	Gamete formation (sperm/egg)
Crossing over	No	Yes (Prophase I)
■ EXAM HACK — Mitosis vs Meiosis trap 'PMAT' for mitosis phase order — Prophase, Metaphase, Anaphase, Telophase. Meiosis = 'Meiosis makes 4' — four haploid daughter cells, NOT two. Crossing over happens ONLY in Prophase I of meiosis — never in mitosis; a favourite trick-question line.		

4. Epithelial Tissue

EPITHELIAL TISSUE — CLASSIFICATION



General Characteristics

- Covers body surfaces, lines cavities/organs/vessels, and forms glands.
- Cells are closely packed with minimal extracellular matrix, joined by junctions (tight, adherens, desmosomes, gap junctions).
- **Avascular** — no blood vessels; nourished by diffusion from underlying connective tissue.
- Sits on a **basement membrane** (basal lamina + reticular lamina) that anchors it to underlying connective tissue.
- Has an apical (free) surface and a basal surface — always polarized.
- Highly regenerative due to a high mitotic rate.

Classification (by shape + number of layers)

Type	Description	Example Location
Simple squamous	Single layer of flat, thin cells	Alveoli, capillary lining (endothelium), lining of body cavities (mesothelium)
Simple cuboidal	Single layer of cube-shaped cells	Kidney tubules, glandular ducts, ovary surface
Simple columnar	Single layer of tall, column-shaped cells; may have microvilli/cilia	Lining of stomach, intestines (with goblet cells)
Pseudostratified columnar	Single layer but nuclei at different levels — looks layered; often ciliated	Respiratory tract (trachea)

Type	Description	Example Location
Stratified squamous	Multiple layers, flattened surface cells; keratinized (skin) or non-keratinized (moist surfaces)	Skin epidermis, oesophagus, mouth
Stratified cuboidal/columnar	Multiple layers of cuboidal/columnar cells (rare)	Large gland ducts (sweat glands)
Transitional	Multiple layers that stretch/change shape (dome to flat)	Urinary bladder, ureters

Glandular Epithelium

- **Exocrine glands:** secrete via ducts onto a surface (e.g. sweat, salivary, sebaceous glands).
- **Endocrine glands:** ductless — secrete hormones directly into the bloodstream (e.g. thyroid, adrenal glands).
- Exocrine secretion modes: **merocrine** (exocytosis, cell intact — e.g. salivary glands), **apocrine** (pinches off part of the cell — e.g. mammary glands), **holocrine** (whole cell ruptures and becomes the secretion — e.g. sebaceous glands).

■ ALZHEIMER'S MODE

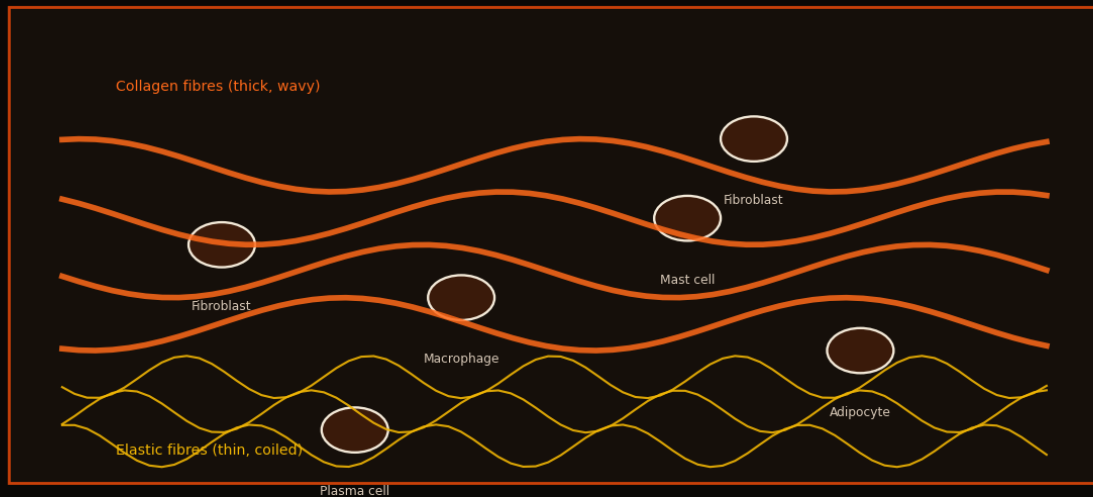
Think of epithelium as the body's 'wallpaper and wrapping paper' — it covers, lines, and wraps every free surface. Simple = one layer = built for fast exchange (thin walls). Stratified = many layers = built for protection (thick walls, like armour).

■ EXAM HACK — Classification trap

Naming rule: shape of the SURFACE (apical) layer names stratified epithelium, but for simple epithelium the shape names the WHOLE single layer. 'Pseudo' stratified is NOT truly layered — all cells touch the basement membrane, only nuclei are staggered. Transitional epithelium is the answer whenever a question mentions 'stretch' + 'bladder'.

5. Connective Tissue Proper

CONNECTIVE TISSUE — GROUND SUBSTANCE + FIBRES + CELLS



General Characteristics

- Most abundant and widely distributed tissue — connects, supports, protects, and binds other tissues together.
- Unlike epithelium: cells are sparse and widely spaced within an abundant **extracellular matrix** (ground substance + fibres).
- Usually well **vascularized** (except cartilage, which is avascular).
- Derived embryologically from mesenchyme (mesoderm).

Cell Types

Cell	Function
Fibroblasts	Most abundant resident cell; synthesize fibres and ground substance
Macrophages	Phagocytose debris/pathogens; derived from monocytes
Mast cells	Release histamine/heparin — mediate inflammation and allergic response
Plasma cells	Derived from B-lymphocytes; secrete antibodies
Adipocytes	Store fat (triglycerides); energy reserve and insulation
Leukocytes	White blood cells that migrate into tissue during inflammation

Fibres of the Matrix

- **Collagen fibres** — thick, strong, flexible but resistant to stretch; provides tensile strength (most abundant protein in the body).
- **Elastic fibres** (elastin) — thin, branching, allow stretch and recoil (e.g. blood vessel walls, lungs).

- **Reticular fibres** — fine collagen (type III) forming delicate supportive networks/stroma for soft organs (liver, spleen, lymph nodes).

Classification

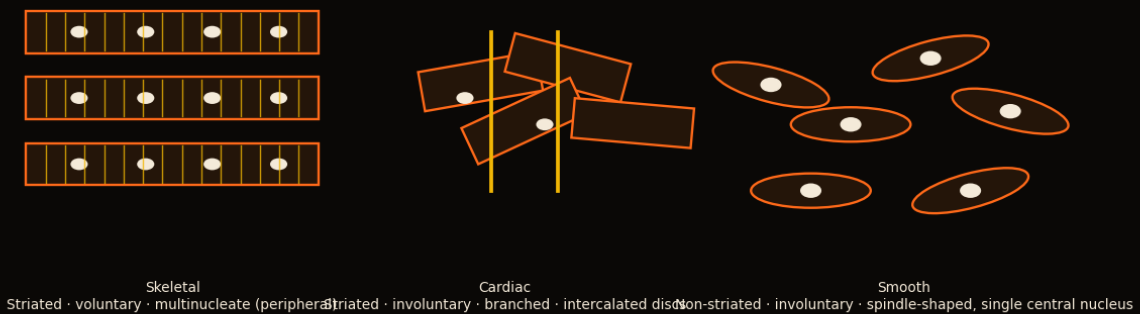
Type	Description	Example
Loose (areolar)	Loosely arranged fibres in abundant ground substance; cushions/supports	Beneath skin, around organs/vessels
Dense regular	Densely packed, parallel collagen fibres — high tensile strength in one direction	Tendons, ligaments
Dense irregular	Densely packed collagen running in multiple directions — resists stress from all directions	Dermis of skin, organ capsules
Adipose	Cells dominated by fat storage (signet-ring appearance)	Subcutaneous fat, around kidneys
Reticular	Fine reticular fibre meshwork supporting free cells	Lymph nodes, spleen, bone marrow stroma

■ EXAM HACK — Fibre recall

Collagen = ROPE (strong, doesn't stretch). Elastin = RUBBER BAND (stretches and snaps back). Reticular = NET (fine supportive mesh for soft organs). Common trap: calling adipose tissue 'just fat storage' — remember it's a true connective tissue with cells (adipocytes) in a matrix.

6. Muscle Tissue

MUSCLE TISSUE — SKELETAL vs CARDIAC vs SMOOTH



General Characteristics

- Specialized for contraction — generates force and movement.
- Key properties: **excitability** (responds to stimuli), **contractility** (shortens), **extensibility** (can be stretched), **elasticity** (recoils).
- Composed of elongated cells called **muscle fibres**, containing contractile proteins actin and myosin.

Feature	Skeletal	Cardiac	Smooth
Appearance	Striated	Striated	Non-striated
Control	Voluntary	Involuntary	Involuntary
Nuclei	Multinucleate (peripheral)	Single/binucleate (central)	Single (central)
Cell shape	Long, cylindrical fibres	Branched, short cells	Spindle-shaped, tapered
Special features	Attached to bone via tendons	Intercalated discs join cells; gap junctions	Found in walls of hollow organs/vessels
Location	Skeletal muscles	Heart wall (myocardium)	GI tract, blood vessels, airways, uterus

Clinical Correlations

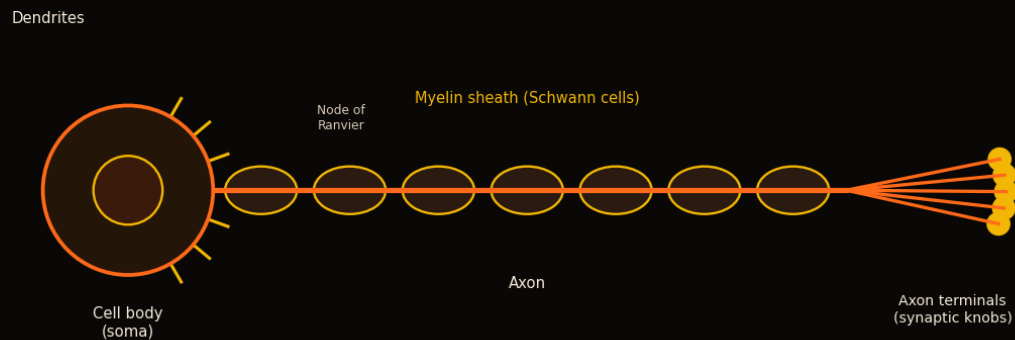
- Skeletal muscle damage can raise serum creatine kinase (CK) — used to detect muscle injury/myocardial infarction.
- Intercalated discs (cardiac muscle) allow electrical signals to spread rapidly — disruption contributes to arrhythmias.
- Smooth muscle dysfunction is implicated in conditions like hypertension (vascular smooth muscle) and irritable bowel syndrome (GI smooth muscle).

■ **EXAM HACK — Muscle ID trap**

'Striated + voluntary' = Skeletal. 'Striated + involuntary' = Cardiac (intercalated discs are the giveaway word).
'Non-striated + involuntary' = Smooth. Multinucleate + peripheral nuclei is ALWAYS skeletal muscle — a classic image-ID question.

7. Nerve Tissue

NEURON — GENERALIZED STRUCTURE



Cell Types

- **Neurons** — the functional/excitable unit; structure: cell body (soma, contains Nissl bodies = rough ER), dendrites (receive signals), axon (conducts impulses away from soma), axon terminals (synapse onto next cell).
- **Neuroglia (glial cells)** — supporting cells, do not conduct impulses but nourish/protect/insulate neurons.

Glial Cell	Location	Function
Astrocytes	CNS	Support neurons, form blood-brain barrier, regulate extracellular environment
Oligodendrocytes	CNS	Myelinate multiple CNS axons
Microglia	CNS	Immune defence — phagocytose debris/pathogens
Ependymal cells	CNS (line ventricles)	Produce/circulate cerebrospinal fluid (CSF)
Schwann cells	PNS	Myelinate ONE PNS axon segment each; aid nerve regeneration
Satellite cells	PNS	Support neuron cell bodies in ganglia

Neuron Classification

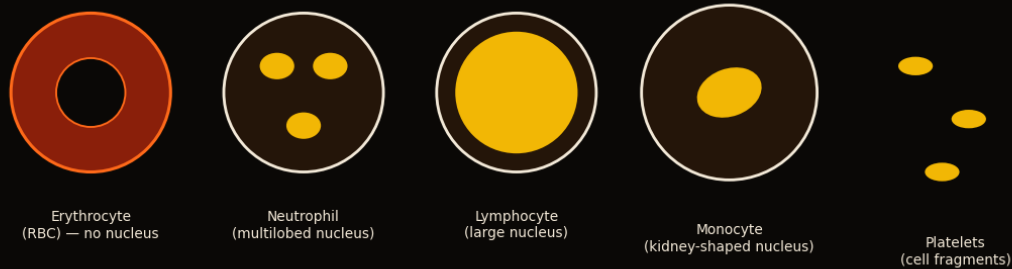
- By structure: **multipolar** (many dendrites, one axon — most common, e.g. motor neurons), **bipolar** (one dendrite, one axon — e.g. retina, olfactory), **unipolar/pseudounipolar** (single process splits — e.g. sensory neurons).
- By function: **sensory (afferent)** — carry impulses toward CNS; **motor (efferent)** — carry impulses away from CNS to effectors; **interneurons** — relay/process signals within the CNS.

■ **EXAM HACK — Myelination trap**

CNS myelin = Oligodendrocytes (one cell, MANY axons). PNS myelin = Schwann cells (one cell, ONE axon segment). Mnemonic: 'Oligo' sounds like 'many' relationships at once; 'Schwann' = one-on-one. Multiple sclerosis attacks CNS myelin (oligodendrocytes); Guillain-Barré attacks PNS myelin (Schwann cells).

8. Blood

BLOOD — FORMED ELEMENTS



- Classified as a fluid connective tissue — cells (formed elements) suspended in a liquid matrix (plasma).
- **Plasma** (~55%): water, plasma proteins (albumin, globulins, fibrinogen), electrolytes, nutrients, wastes, hormones.
- **Formed elements** (~45%): erythrocytes, leukocytes, platelets — all arise from haematopoietic stem cells in red bone marrow.

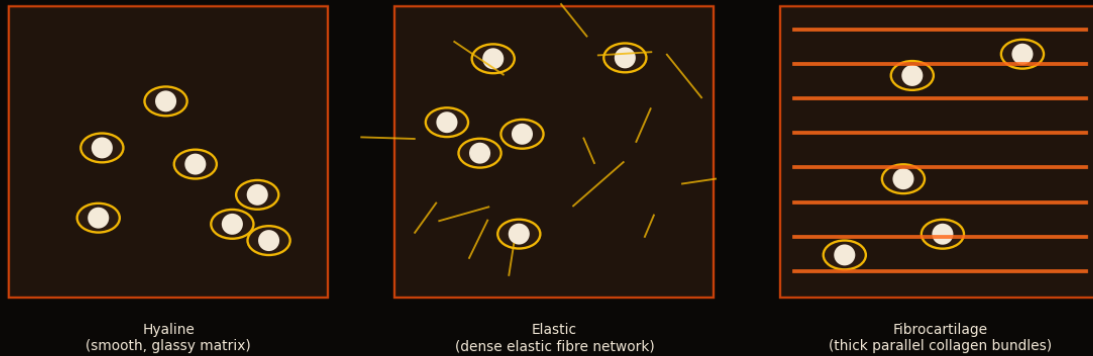
Cell	Key Feature	Function
Erythrocytes (RBCs)	Biconcave disc, no nucleus, contain haemoglobin	Transport O ₂ and CO ₂
Neutrophils	Multilobed nucleus, granular cytoplasm; most abundant WBC	First responders — phagocytose bacteria
Eosinophils	Bilobed nucleus, red/orange granules	Combat parasites, allergic reactions
Basophils	Large dark granules, obscure nucleus; least common WBC	Release histamine/heparin (inflammation)
Lymphocytes	Large nucleus, thin rim of cytoplasm	B cells (antibodies), T cells (cell-mediated immunity)
Monocytes	Largest WBC, kidney-shaped nucleus	Become macrophages — phagocytosis
Platelets (thrombocytes)	Cell fragments, no nucleus	Blood clotting (haemostasis)

■ EXAM HACK — WBC recall order

'Never Let Monkeys Eat Bananas' = Neutrophils, Lymphocytes, Monocytes, Eosinophils, Basophils = descending order of abundance. Trap: platelets are NOT true cells — they're cytoplasmic fragments of megakaryocytes, and have no nucleus.

9. Cartilage

CARTILAGE — HYALINE vs ELASTIC vs FIBROCARILAGE



- A specialized, resilient, semi-rigid connective tissue that resists compression while allowing some flexibility.
- **Avascular** — nourished by diffusion; heals slowly.
- Cells: **chondroblasts** (immature, secrete matrix) mature into **chondrocytes**, housed in small cavities called **lacunae**.
- Surrounded (except at joint surfaces) by a fibrous **perichondrium**, which supplies nutrients and allows appositional growth.

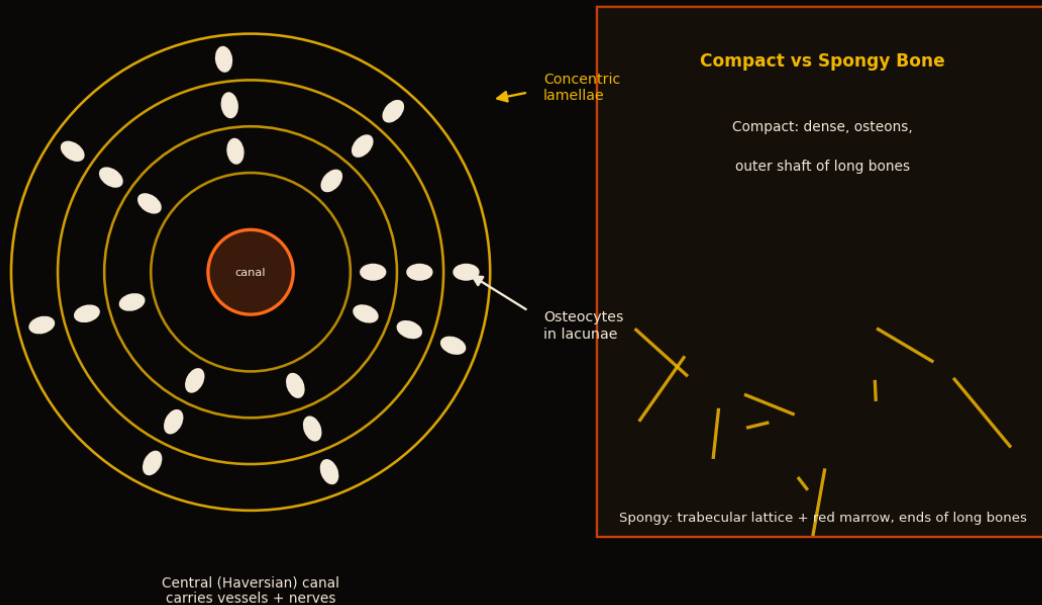
Type	Matrix Features	Location	Notes
Hyaline	Glassy, smooth, fine collagen fibres	Articular surfaces, trachea, nasal septum, costal cartilage, fetal skeleton	Most common type
Elastic	Dense elastic fibre network	External ear (pinna), epiglottis	Provides flexible support
Fibrocartilage	Thick bundles of collagen fibres	Intervertebral discs, pubic symphysis, menisci	Strongest — resists heavy compression/shock

■ EXAM HACK — Cartilage ID trap

Hyaline = 'glassy/glass-like' — the **DEFAULT** answer whenever a question just says 'cartilage' without more detail. Fibrocartilage **NEVER** has a perichondrium — it blends directly into surrounding dense connective tissue, a favourite trick point.

10. Bone

BONE — OSTEON (HAVERSIAN SYSTEM)



General Characteristics

- A specialized, rigid connective tissue — matrix is calcified (hydroxyapatite crystals: calcium + phosphate) giving it hardness.
- Highly **vascularized** and capable of self-repair, unlike cartilage.
- Functions: support, protection, movement (lever for muscles), mineral storage (Ca^{2+} , PO_4^{3-}), and haematopoiesis (red marrow).

Bone Cells

Cell	Function
Osteoprogenitor cells	Stem cells that differentiate into osteoblasts
Osteoblasts	Bone-BUILDING cells — synthesize osteoid (unmineralized matrix)
Osteocytes	Mature osteoblasts trapped in lacunae — maintain bone matrix, sense mechanical stress
Osteoclasts	Large, multinucleate cells — bone-RESORBING (break down matrix); derived from monocyte lineage

Microscopic Structure

- **Compact (cortical) bone:** dense, organized into structural units called **osteons (Haversian systems)** — concentric lamellae around a central (Haversian) canal carrying blood vessels/nerves; Volkmann's canals connect adjacent osteons.
- **Spongy (cancellous) bone:** lattice of thin plates called **trabeculae**, with spaces filled by red bone marrow — lighter, found at the ends of long bones and within flat bones.

- **Periosteum:** fibrous membrane covering the outer surface of bone (except at joints) — site of appositional growth and tendon/ligament attachment.
- **Endosteum:** thin membrane lining the internal marrow cavity.

Bone Formation (Ossification)

- **Intramembranous ossification:** bone forms directly from mesenchyme (no cartilage model) — flat bones of the skull, clavicle.
- **Endochondral ossification:** a hyaline cartilage model is gradually replaced by bone — most bones of the body, including long bones (growth continues at the epiphyseal/growth plate until maturity).

■ ALZHEIMER'S MODE

Osteoblast = 'B' for Build. Osteoclast = 'C' for Chew/Chomp (breaks bone down). The osteon is like a tree trunk's growth rings — concentric layers (lamellae) wrapped around a central pipe (Haversian canal) carrying the blood supply.

■ EXAM HACK — Bone trap

Osteocytes are TRAPPED, mature, maintenance-only cells — they do not divide or actively build new matrix (osteoblasts do that). Growth plate (epiphyseal plate) is CARTILAGE, not bone — a classic 'which of these is NOT ossified' trick question.

End of Compendium

This compendium consolidates every ANA203 topic taught to date: The Cell, Membrane Transport, Cell Division, Epithelial Tissue, Connective Tissue Proper, Muscle Tissue, Nerve Tissue, Blood, Cartilage, and Bone. Pair this with the ANA203 CBTs on the INDIGENE Study Console for active recall practice.